

PRD: Car Insurance Claims Agent

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Context & Vision

We are working with a leading car insurance company to help them improve their claims process through an agentic workflow that increases accuracy and reduces costs.

Approximately 40 million auto insurance claims are reported and processed in the United States every year. By some reports, it represents 70-80% of the operating cost of vehicle insurance companies, which have an aggregate market of \$2.27 trillion globally. Solutions for this flow are costly for this space as well, with the global market for claims processing software estimated at about \$47B. The size of this market alone and the scale of our customer creates the opportunity for significant impact.

Our vision is to **redefine the claims experience for the agentic era** by delivering more human claims experiences and increasing the speed, accuracy, and efficiency of the claim management process.

Objectives & Metrics

Our objectives for this feature are to increase the efficiency of claims processing and deliver better customer results, as measured by:

1. **Time-to-Resolution:** the average time it takes to close a claim
2. **Number of Claims Processed**
3. **Cost per Claim Processed**
4. **NPS** of customers submitting claims

We can also measure a **series of evals** on the agent, including latency, hallucination rate, linting for structured responses and accuracy of tool calls.

Risks & Ethics

While we are measuring positive improvements to the claims process, we have to keep in mind a few trade-offs when implementing agentic systems:

1. **Abuse or fraud:** It's possible for bad actors to abuse an autonomous system for insurance fraud; these claims should be denied before reaching an adjuster

2. **Risk of “unreasonable investigations” or bias:** At the same time, insurers cannot deny a car insurance claim arbitrarily. It will be important to monitor potential systematic (e.g., regional or economic) biases in our agent.
3. **False positives or errors increasing burden on senior claims adjusters**
4. **Economic displacement of frontline claims adjusters:** we can consider how our agent could upskill frontline adjusters to be more effective reviewers of claims.

Persona

For our initial MVP, we'll focus on building an agent to process personal auto insurance claims because they represent the [vast majority](#) of claims in the US. We'll also focus exclusively on collision claims (damage to the vehicle) vs. other types of claims (e.g., liability or bodily injury).

User Stories / Requirements

P0 ▾ Users can initiate a claim from a logged-in session

P0 ▾ Users can input required information for insurance claims in deterministically captured fields:

- date, location, description of the accident, police department, opposite party
- Fields should be validated if they are critical components to the claims process

P0 ▾ Users can upload documents related to the claim

- Images of Damage
- Police Report Information

P0 ▾ After submitting initial claim information, users are presented with a chat interface to interact with the agent and view initial results. The **agent architecture** should include:

- **An Orchestrator** that reasons and routes tool calls; gauges intent and constrains action and responses to the claims process
- A **Damage Assessment** tool that calls a computer vision tool to review the documents and images, and returns structured output
- Tool 3: Anti-fraud Reviewer
- An **Estimate Calculator** that takes the structured output of the Damage Assessment and provides a potential estimate for the claim
- A **Repair Shop Tool** to provide the user with actionable next steps following their claim

P0 ▾ Human in the loop: After submitting initial claim information, users should validate the initial damage assessment provided by the agent

P0 ▾ After submitting the information the user should be asked to wait while a senior claims adjuster reviews the claim.

P0 ▾ If approved, the user is given a claim number as well as an insurer approved repair shop in the user's location

P0 ▾ Guardrails (e.g., another agent), are used to ensure the agent stays on task

- **Fraud:** Another agent should monitor the claim for potential fraud and flag to the senior claims adjuster or deny outright
- **A task guardrail,** to ensure that the agent stays within the claims process and does not answer unrelated queries

P0 ▾ If not approved, or pending further review, users are notified that their claim is still pending. Users are asked to check status of their claim on a dashboard or reach out to support

Design Principles

- The intake of initial details is formal (and not chat-based) due to the need for robust data input and the cost of LLM iterating to select key information
- We anchored on clarity of tool calls in the chat interface to illustrate progress
- Embedded in the chat UX is the ability for the user to validate the agent's damage outputs in order to build more trust in the agent and give a sense of control
- When final approval is confirmed by the adjuster, an explanation of what was approved is presented to the user for information clarity

Notes on prioritization / what we're not doing

1. Guardrails were included as P0 in the PRD due to enterprise risk of non-deterministic systems
2. We've deprioritized commercial insurance claims, reducing complexity. We also deprioritized other claim types (e.g., Liability, bodily injury)
3. We deprioritized an assistant for the claims adjuster, but this information would be similarly useful for an agentic pair-adjuster for the human-in-the-loop
4. We deprioritized authentication in the agentic experience by assuming the user is logged in when filing their claim